

Fuzzification

- Fuzzification is the first step in a fuzzy logic system, where crisp input values are converted into fuzzy values, enabling the system to handle the inherent uncertainty and vagueness of real-world data.
- In simple terms, fuzzification takes clear, well-defined numbers (like 35°C) and maps them into fuzzy sets or categories (such as “warm” or “hot”).

Steps in Fuzzification:

Input Values:

Fuzzification starts with real, crisp inputs (e.g., temperature, speed, height, etc.) that are measured from the environment or system.

Membership Functions:

A membership function (MF) defines how each input value corresponds to a fuzzy set. A fuzzy set is a collection of elements, each of which has a degree of membership between 0 and 1.

Fuzzy Sets

- Fuzzy sets represent categories or linguistic variables. These sets could represent concepts like “low,” “medium,” “high,” or more specific terms like “very hot,” “cold,” etc.
- The fuzzification process calculates the degree to which each crisp input belongs to each fuzzy set.

Defuzzification

- Defuzzification is the inverse of fuzzification. It involves converting fuzzy output values back into a single crisp value to make actionable decisions.

Centroid Method (Center of Area Method):

The centroid method is the most widely used defuzzification technique. It involves finding the center of gravity (or the “centroid”) of the area under the curve defined by the fuzzy output.

Process:

- The fuzzy set is represented by a curve that reflects the membership degrees.
- The centroid (center of area) is calculated, which gives the crisp output by averaging the “weighted” positions of the fuzzy set elements.
- This is essentially the center of mass of the fuzzy set.

Mean of Maxima (MOM)

The Mean of Maxima (MOM) method is based on finding the crisp value corresponding to the maximum membership degree (the peak of the fuzzy set).

Process:

- Identify the highest membership value in the fuzzy set.
- If there are multiple points with the same highest membership value, take the average of those points to get the crisp output.

Weighted Average Method

- The Weighted Average Method involves calculating a weighted average of the output fuzzy sets. Each fuzzy set has a corresponding crisp output, and these outputs are weighted by their membership degrees.
- **Process:**
 - For each fuzzy output set, calculate its centroid or a representative crisp value.
 - Multiply each crisp value by the degree of membership in the fuzzy set.
 - Sum these products and divide by the sum of the membership degrees.

Fuzzy Rule Based Systems

General Structure of FRBS:

A fuzzy rule represents knowledge

IF <condition> THEN <action>

or

IF antecedent THEN consequent

Canonical Form of Fuzzy Rules

A rule-based system in canonical form expresses each rule as:

IF (condition1) AND (condition2) AND ... THEN (action/conclusion)

Formally:

IF x_1 is A_{i1} AND x_2 is A_{i2} AND ... AND x_n is A_{in} THEN y is B_i^{**}

Where:

$x_1, x_2, \dots, x_n$ = input variables

$A_{i1}, A_{i2}, \dots, A_{in}$ = fuzzy sets/conditions

y = output variable

B_i = output fuzzy set

i = rule number

Canonical Form of Fuzzy Rules

- A complete fuzzy rule base in canonical form:

$$\mathcal{R} = \{R_1, R_2, \dots, R_M\}$$

Each rule:

$$R_i: \text{ IF } (x_1 \in A_{i1}) \wedge (x_2 \in A_{i2}) \wedge \dots \wedge (x_n \in A_{in}) \text{ THEN } y \in B_i$$

$$i = 1, 2, \dots, M$$

Canonical Form of Fuzzy Rules

In fuzzy systems, canonical form ensures:

- **All rules have the same number of antecedents**

Even if a rule doesn't need a condition, we use a neutral condition like "x is ANY".

- **All antecedents are connected by AND**

OR can be transformed into AND using fuzzy logic combinations.

- **Each rule produces a single output**

Multiple Conjunctive Antecedents (AND)

Rule states:

IF x is A1 AND x is A2 AND x is A3 THEN y is B

Equivalent to:

- Create a combined fuzzy set $A_{\text{new}} = A1 \cap A2 \cap A3$
- Using membership:

$$\mu_{A_{\text{new}}}(x) = \min(\mu_{A1}(x), \mu_{A2}(x), \mu_{A3}(x))$$

Multiple disjunctive antecedents (OR)

Rule stated:

IF x is A1 OR x is A2 THEN y is B

Equivalent to:

$$A_{\text{new}} = A1 \cup A2$$

$$\mu_{A_{\text{new}}}(x) = \max(\mu_{A1}(x), \mu_{A2}(x))$$

Aggregation of Fuzzy Rules

When multiple rules operate simultaneously, their results must be combined.

Conjunctive rule system (AND system)

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If all rules must hold:

$$Y = Y_1 \cap Y_2 \cap \dots \cap Y_n$$

$$\mu_Y(y) = \min(\mu_{Y_1}(y), \dots, \mu_{Y_n}(y))$$

Used in **strict control systems**.

Disjunctive rule system (OR system)

-

If at least one rule is enough:

$$Y = Y_1 \cup Y_2 \cup \dots \cup Y_n$$

$$\mu_Y(y) = \max(\mu_{Y_1}(y), \dots, \mu_{Y_n}(y))$$

Used in **decision-making**.

Linguistic Variables

A linguistic variable is a fuzzy variable.

For example, the statement “John is tall” implies that the linguistic variable *John* takes the linguistic value *tall*.

In fuzzy expert systems, linguistic variables are used in fuzzy rules.

For example:

IF wind is strong THEN sailing is good

IF project_duration is long THEN completion_risk is high

IF speed is slow THEN stopping_distance is short

Linguistic Variables

- Variables in mathematics normally take numeric values, although non-numeric linguistic variables are frequently employed in fuzzy logic to make the expression of rules and facts easier.
- Linguistic variables are variables with a value made up of linguistic concepts (also known as linguistic words) rather than numbers

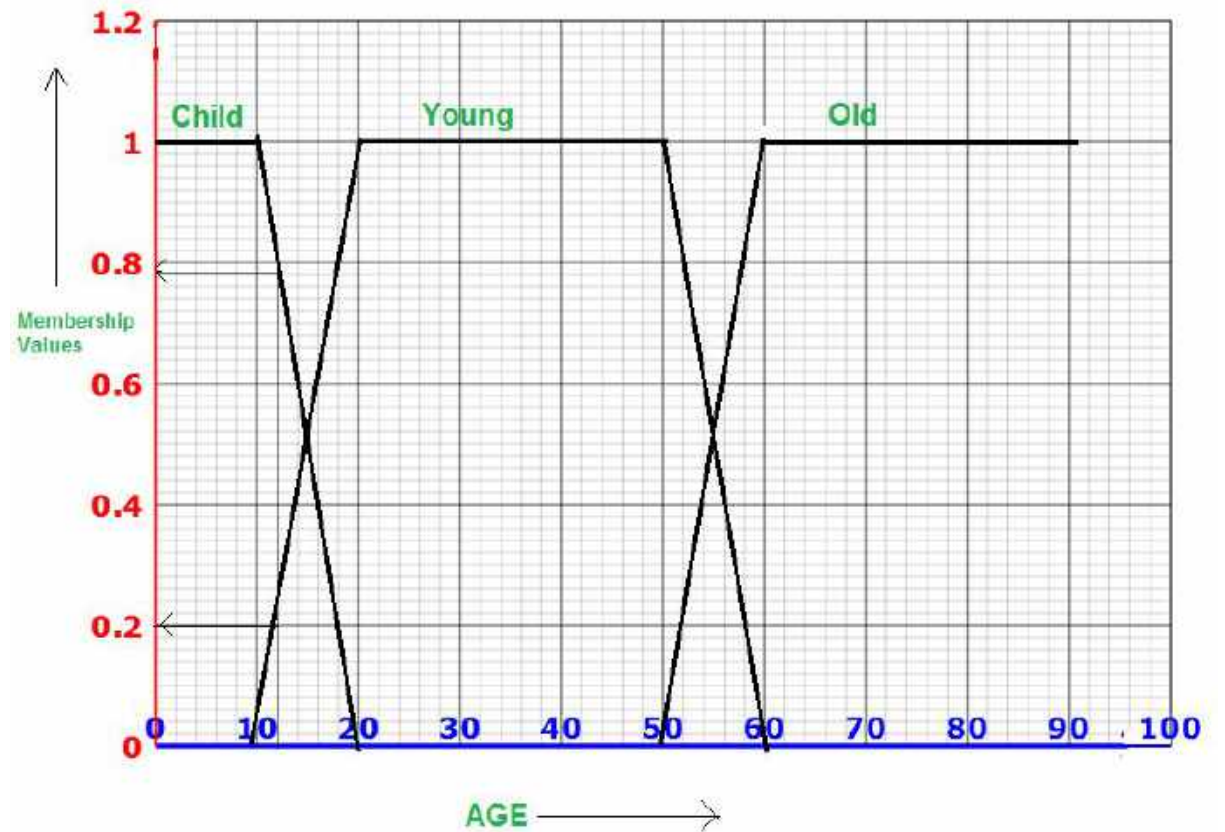
Linguistic Variables

Example:

$\text{AGE} = \{\text{Child}, \text{Young}, \text{Old}\}$

Each AGE linguistic phrase has a membership function for a specific age range.

These membership values can then be used to identify whether a person is a child, a young person, or an elderly person.



Linguistic Variables

The range of possible values of a linguistic variable represents the universe of discourse of that variable.

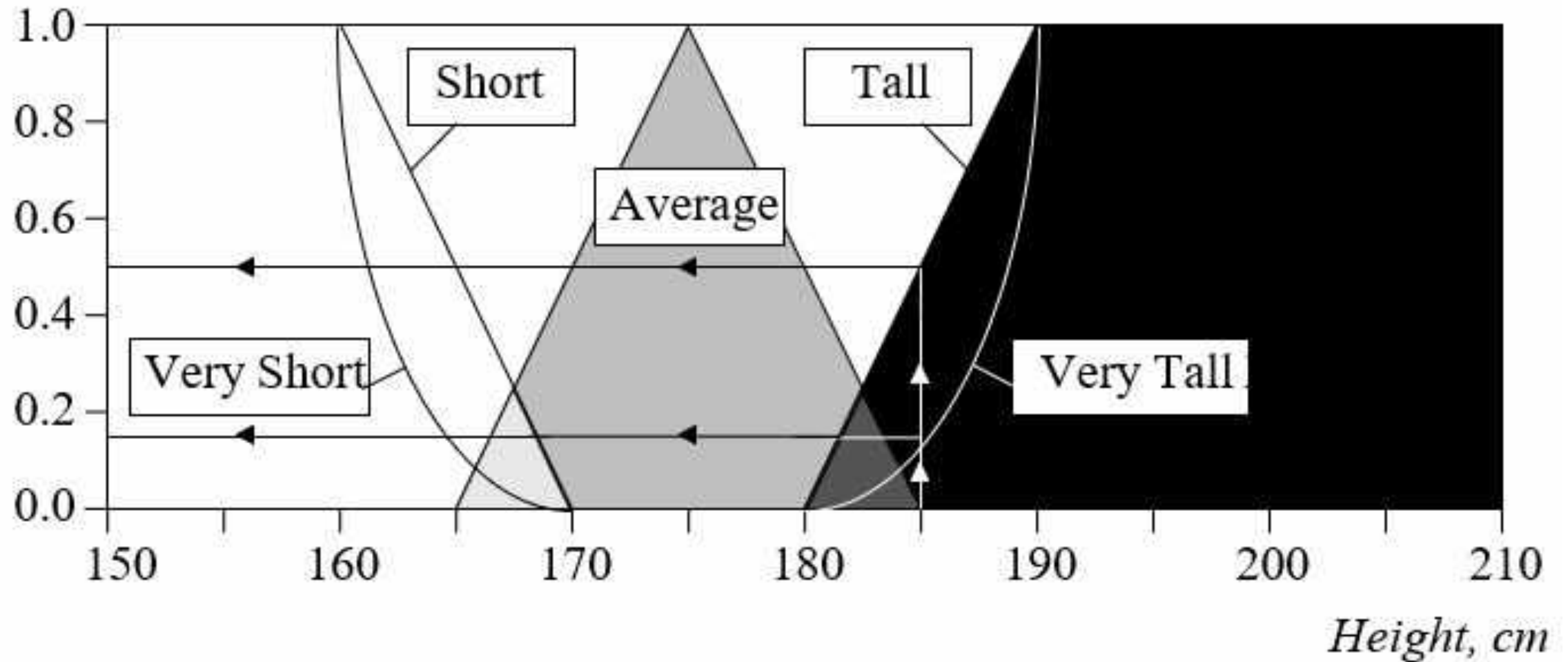
For example, the universe of discourse of the linguistic variable *speed* might have the range between 0 and 220 km/h and may include such fuzzy subsets as *very slow*, *slow*, *medium*, *fast*, and *very fast*.

A linguistic variable carries with it the concept of fuzzy set qualifiers, called *hedges*.

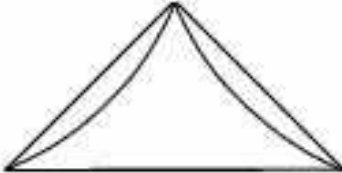
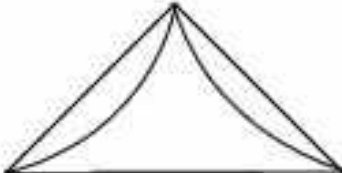
Hedges are terms that modify the shape of fuzzy sets. They include adverbs such as *very*, *somewhat*, *quite*, *more or less* and *slightly*.

Fuzzy sets with the hedge very

Degree of Membership



Representation of hedges in fuzzy logic

<i>Hedge</i>	<i>Mathematical Expression</i>	<i>Graphical Representation</i>
A little	$[\mu_A(x)]^{1.3}$	
Slightly	$[\mu_A(x)]^{1.7}$	
Very	$[\mu_A(x)]^2$	
Extremely	$[\mu_A(x)]^3$	

Fuzzy Extension Principle

Definition: Let $X = X_1 \times X_2 \times \dots \times X_n$ be the cartesian product of the universes X_i , $i = 1, \dots, n$, and $\tilde{A}_i \subset X_i$ be fuzzy sets in X_i , and let $f : X \rightarrow Y$ be a function (a mapping) such that $y = f(x_1, x_2, \dots, x_n)$, $x_i \in X_i$, $i = 1, \dots, n$, and Y is also an universe.

The extension principle permits the definition of a fuzzy set $\tilde{B} \subset Y$, $\tilde{B} = \{(y, \mu_{\tilde{B}}(y)) \mid y = f(x_1, x_2, \dots, x_n), (x_1, x_2, \dots, x_n) \in X\}$, where:

$$\mu_{\tilde{B}}(y) = \begin{cases} \sup_{x \in f^{-1}(y)} \min(\mu_{\tilde{A}_1}(x_1), \dots, \mu_{\tilde{A}_n}(x_n)), & \text{if } (\exists) f^{-1}(y) \\ 0, & \text{otherwise} \end{cases}$$

where $x = (x_1, x_2, \dots, x_n)$

Fuzzy Extension Principle - Explanation

In the general form, the universe X is the cartesian product of the universes $X_1, X_2, \dots, X_n$, and the fuzzy set $\tilde{A}$ is the cartesian product of the fuzzy sets $\tilde{A}_1, \tilde{A}_2, \dots, \tilde{A}_n$, i.e.

$$\tilde{A} = \tilde{A}_1 \times \tilde{A}_2 \times \dots \times \tilde{A}_n$$

According to the formula for the cartesian product, it results that $\mu_{\tilde{A}}(x) = \min(\mu_{\tilde{A}_1}(x_1), \dots, \mu_{\tilde{A}_n}(x_n))$, where $x = (x_1, x_2, \dots, x_n)$

Of course, the extension principle in the reduced form can be obtained from the extension principle in the general form, by making $n = 1$.

Fuzzy Extension Principle - Example

Let $X = Y = \mathbb{Z}$, the set of integer numbers, let $\tilde{A} \subset \mathbb{Z}$, $\tilde{A} = \{(-2, 0.3), (-1, 0.5), (0, 0.8), (1, 1), (2, 0.7), (3, 0.1)\}$ and let the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x^2$. Which is the fuzzy set $\tilde{B} \subset \mathbb{Z}$ with $\tilde{B} = f(\tilde{A})$ (i.e. $\tilde{B}$ is the image of the set $\tilde{A}$ through the function f) ?.

- ▶ $\mu_{\tilde{B}}(0) = \mu_{\tilde{A}}(0) = 0.8$ and $\mu_{\tilde{B}}(9) = \mu_{\tilde{A}}(3) = 0.1$
- ▶ $\mu_{\tilde{B}}(1) = \sup_{x \in f^{-1}(1)}(\mu_{\tilde{A}}(x)) = \sup(\mu_{\tilde{A}}(-1), \mu_{\tilde{A}}(1)) = \sup(0.5, 1) = 1$
- ▶ $\mu_{\tilde{B}}(4) = \sup_{x \in f^{-1}(4)}(\mu_{\tilde{A}}(x)) = \sup(\mu_{\tilde{A}}(-2), \mu_{\tilde{A}}(2)) = \sup(0.3, 0.7) = 0.7$
- ▶ If we consider that all the integer numbers between 0 and 9 belong to the set $\tilde{B}$, then: $\mu_{\tilde{B}}(2) = \mu_{\tilde{B}}(3) = \mu_{\tilde{B}}(5) = \mu_{\tilde{B}}(6) = \mu_{\tilde{B}}(7) = \mu_{\tilde{B}}(8) = 0$ because 2, 3, 5, 6, 7, 8 are not the image of any element from $\tilde{A}$ through the function f .
- ▶ In conclusion, $\tilde{B} = \{(0, 0.8), (1, 1), (2, 0), (3, 0), (4, 0.7), (5, 0), (6, 0), (7, 0), (8, 0), (9, 0.1)\}$

Fuzzy Inference System

- A Fuzzy Inference System (FIS) is a system that uses fuzzy logic to map inputs to outputs. Unlike traditional binary logic systems (where values are either true or false), fuzzy logic allows for a degree of truth or falsehood, with values ranging between 0 and 1.
- FIS is used in various applications like control systems, decision-making processes, and pattern recognition.
- The basic concept of FIS involves taking fuzzy input values, processing them through a series of rules and an inference engine, and then producing fuzzy output values.

Fuzzy Inference System

A typical FIS consists of four main components:

- **Fuzzification:** Converts crisp inputs (real-world values) into fuzzy values.
- **Rule Base:** Contains a set of if-then rules that describe how the inputs relate to the output.
- **Inference Engine:** Processes the fuzzy input using the rules from the rule base to produce fuzzy output.
- **Defuzzification:** Converts the fuzzy output into a crisp value, which is the final output.

Fuzzification (Input)

This converts the crisp (precise) input values into fuzzy values, typically represented as membership functions.

For example, if the input is the temperature in a room, we might fuzzify the temperature into categories like “cold,” “warm,” and “hot” with values like:

Cold: 0 to 20°C

Warm: 15 to 30°C

Hot: 25 to 40°C

The fuzzification process involves determining the degree to which a crisp value belongs to each category. For instance, a temperature of 18°C might belong to “cold” by 0.3, to “warm” by 0.7, and not belong to “hot” at all.

Rule Base

- The rule base is a collection of “if-then” rules that define the system’s behavior. These rules map fuzzy inputs to fuzzy outputs.

A typical if-then rule might look like:

If temperature is cold, then fan speed is low

If temperature is warm, then fan speed is medium

If temperature is hot, then fan speed is high

- In FIS, these rules are generally expressed in terms of fuzzy sets and operations (like AND, OR, NOT). The combination of rules creates a system of behavior based on fuzzy logic.

Inference Engine

- The inference engine is the component responsible for processing the fuzzy inputs using the rule base and producing fuzzy outputs. It follows the fuzzy reasoning process, which combines the fuzzy input values with the rules in the rule base.
- There are different methods of fuzzy inference used by the engine, like **Mamdani** and **Sugeno**, which will be explained in more detail shortly.

Inference Engine

The inference engine works by:

- **Evaluating the rule base:** It checks which rules apply based on the fuzzy inputs.
- **Aggregating the rule outputs:** For each active rule, the output fuzzy sets are calculated.
- **Fuzzy logic operations:** It performs operations (like AND, OR, or Min/Max) to combine the fuzzy outputs from different rules.

Defuzzification

Defuzzification is the process of converting the fuzzy output into a crisp (precise) value, which is the final output of the system. This step is essential to transform fuzzy results into something actionable or usable in real-world applications.

Mamdani Method

- Mamdani's fuzzy inference method is the most commonly seen fuzzy methodology. Mamdani's method was among the first control systems built using fuzzy set theory.
- It was proposed in 1975 by Ebrahim Mamdani as an attempt to control a steam engine and boiler combination by synthesizing a set of linguistic control rules obtained from experienced human operators.

Mamdani's Method

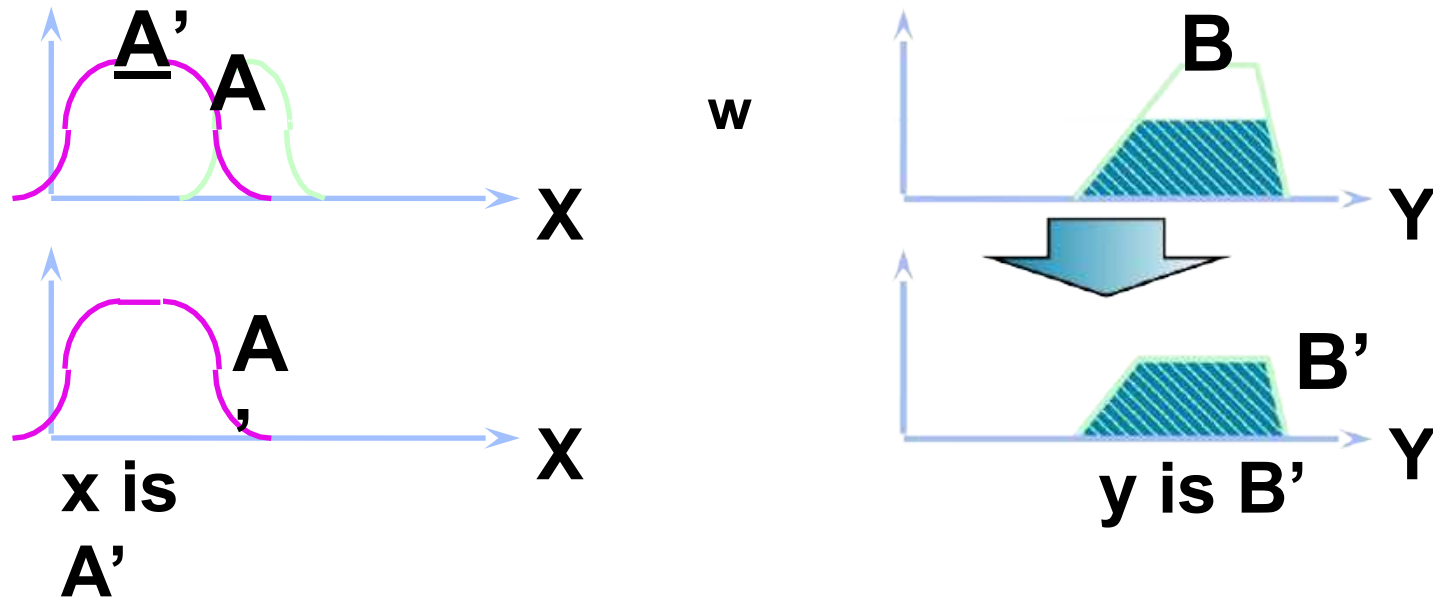
To compute the output of this FIS given the inputs, one must go through six steps:

1. Determining a set of fuzzy rules
2. Fuzzifying the inputs using the input membership functions,
3. Combining the fuzzified inputs according to the fuzzy rules to establish a rule strength,
4. Finding the consequence of the rule by combining the rule strength and the output membership function,
5. Combining the consequences to get an output distribution,
6. Defuzzifying the output distribution (this step is only if a crisp output (class) is needed).

Fuzzy Reasoning (Fuzzy Input)

Single rule with single antecedent Rule: if x is A then y is B Fact: x is A'

Conclusion: y is B' Graphic Representation:

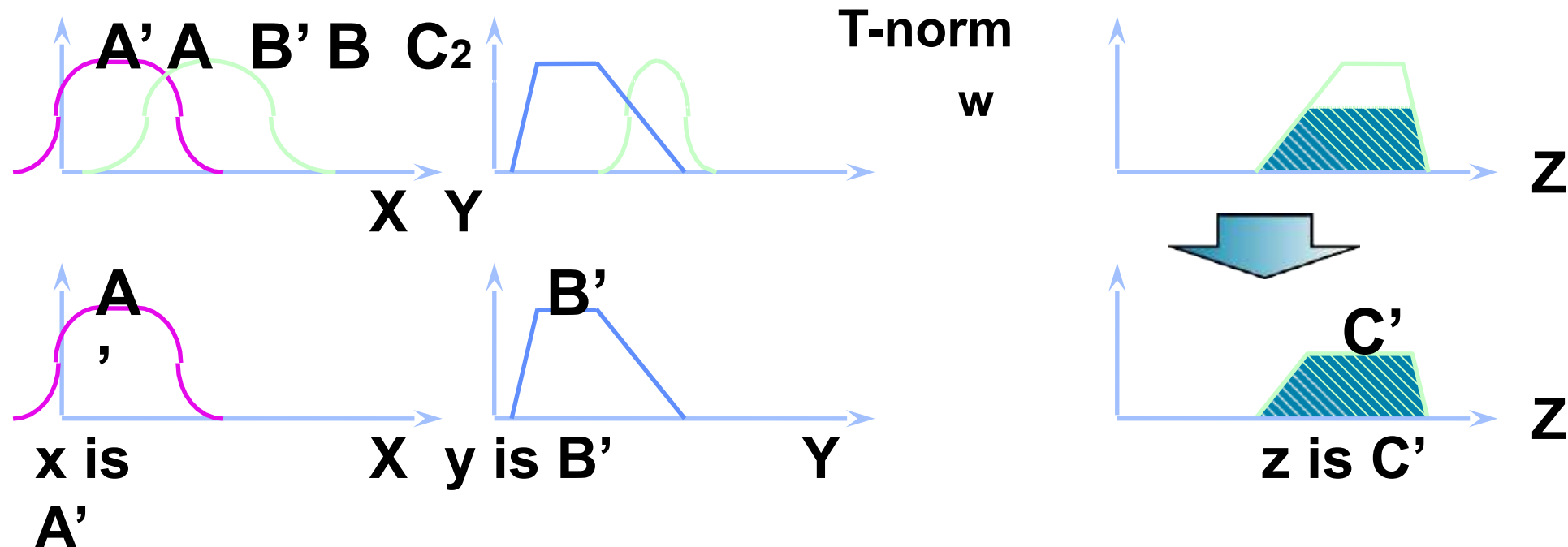


Fuzzy Reasoning (Two Fuzzy Inputs)

Single rule with multiple antecedent Rule: if x is A and y is B then z is C Fact: x is A' and y is B'

Conclusion: z is C'

Graphic Representation:



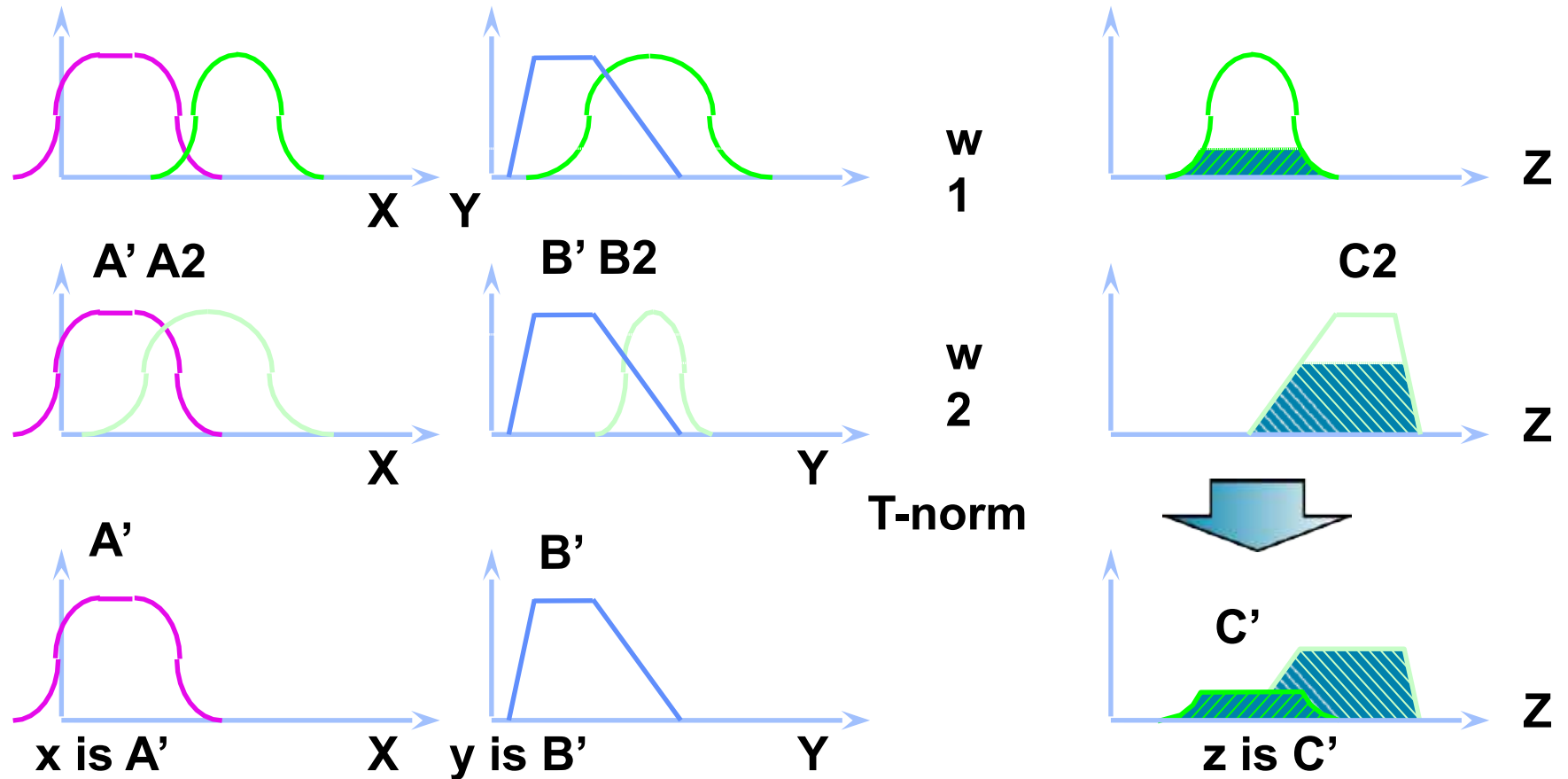
Fuzzy Reasoning (Two Fuzzy Inputs)

Multiple rules with multiple antecedent

Rule 1: if x is A1 and y is B1 then z is C1

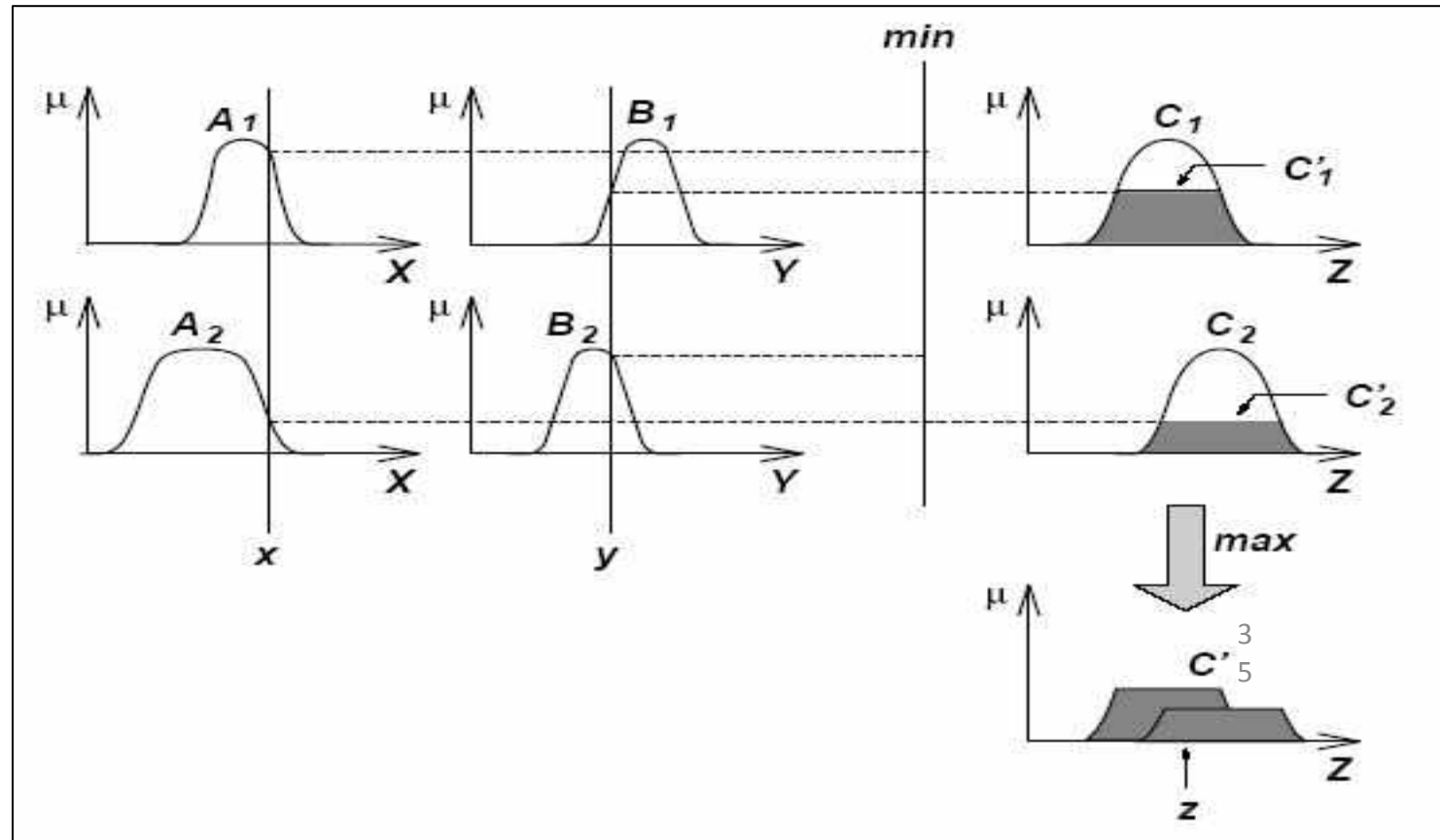
Rule 2: if x is A2 and y is B2 then z is C2
Fact: x is A' and y is B'

Conclusion: z is C'



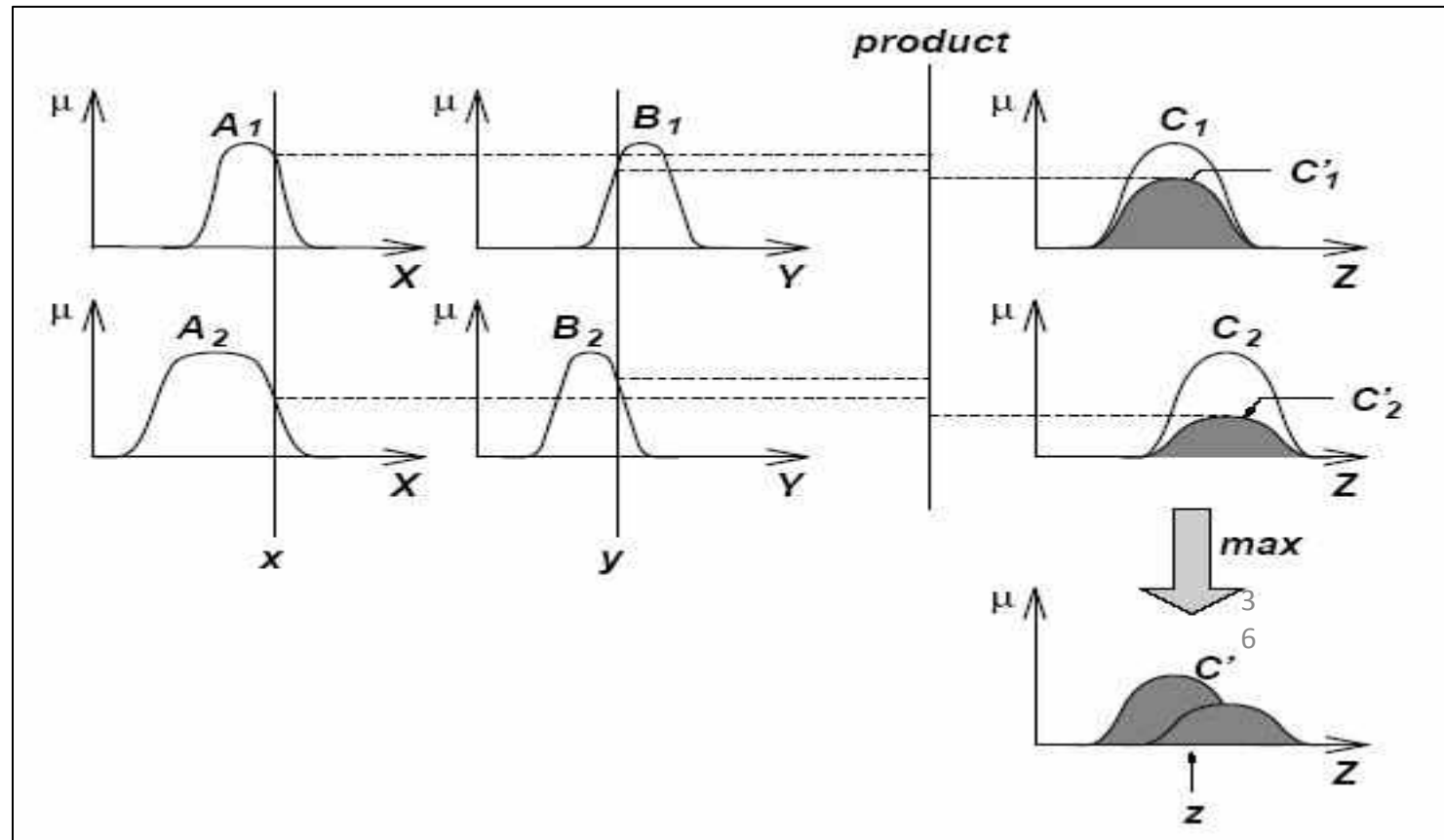
The Reasoning Scheme (Two Crisp Inputs)

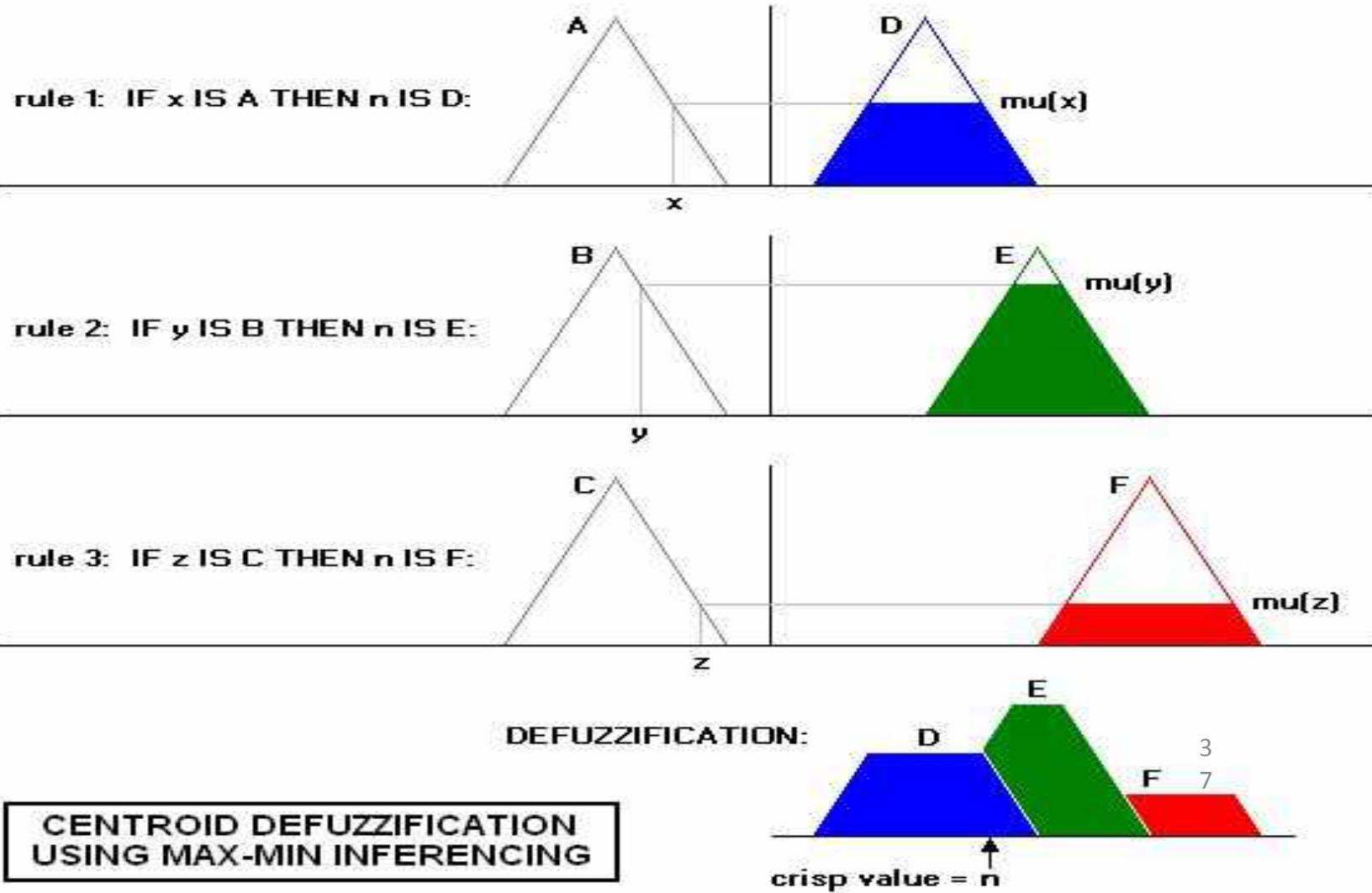
Max-Min Composition is used.



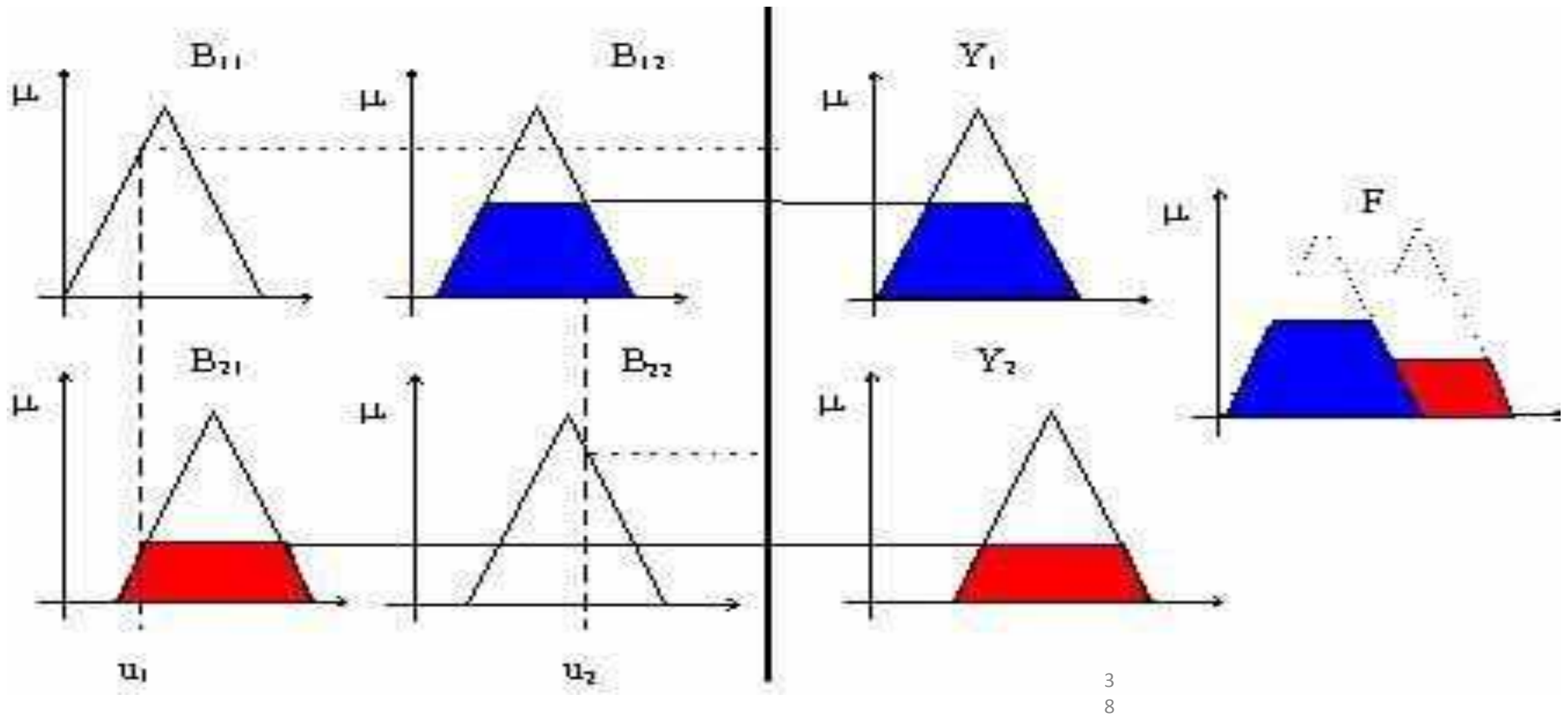
The Reasoning Scheme (Two Crisp Inputs)

Max-Product Composition is used.





Mamdani composition of three SISO fuzzy outputs

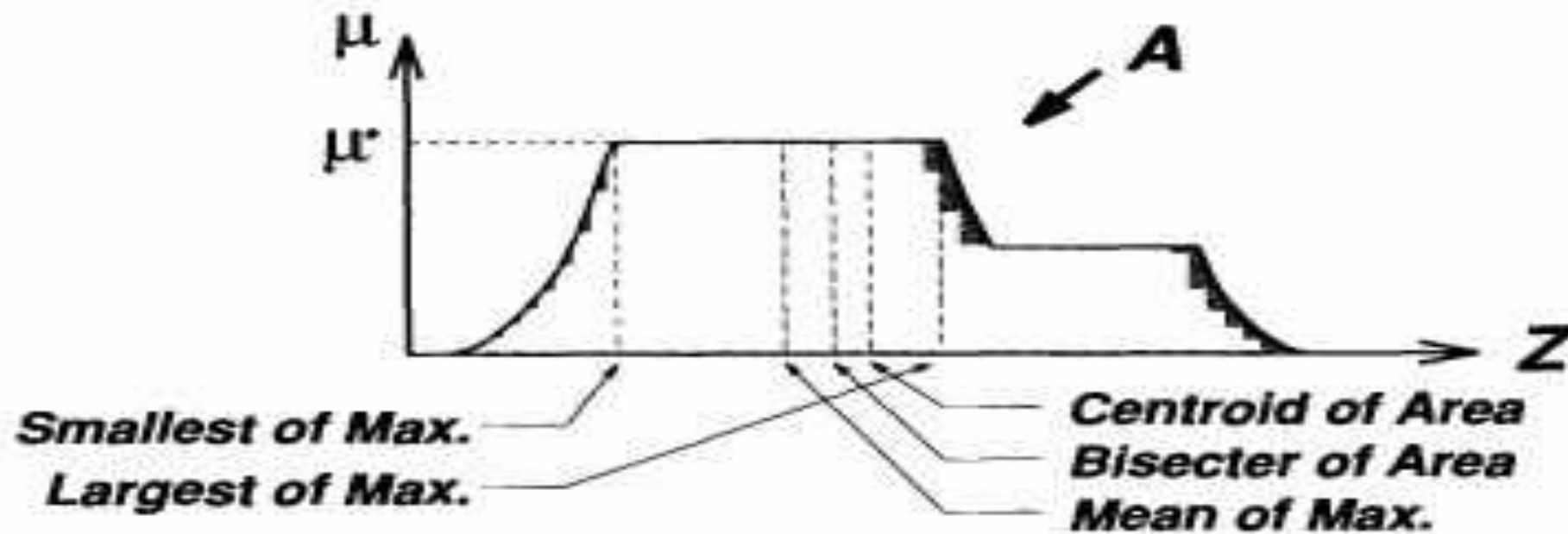


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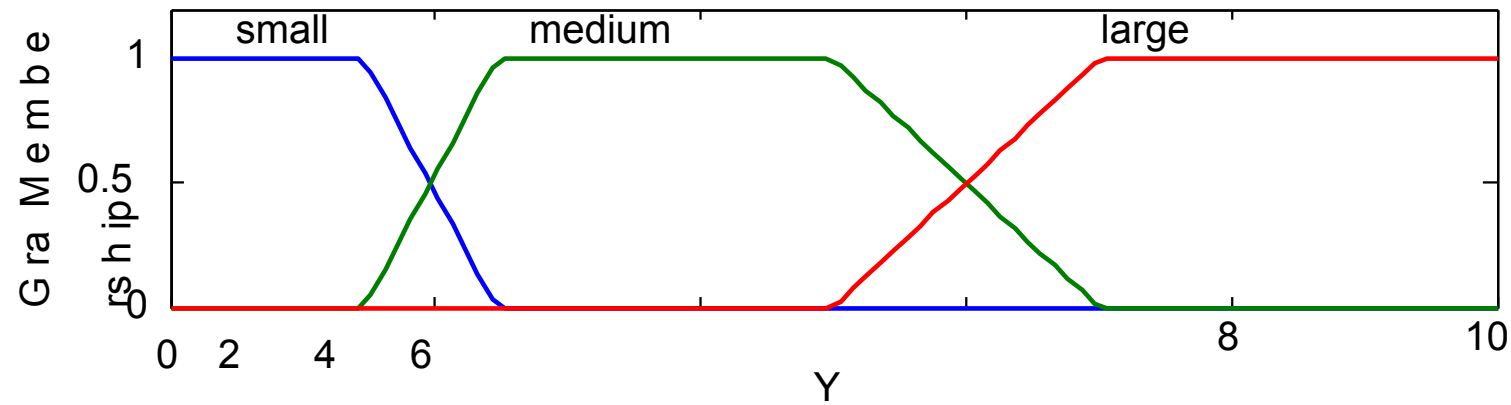
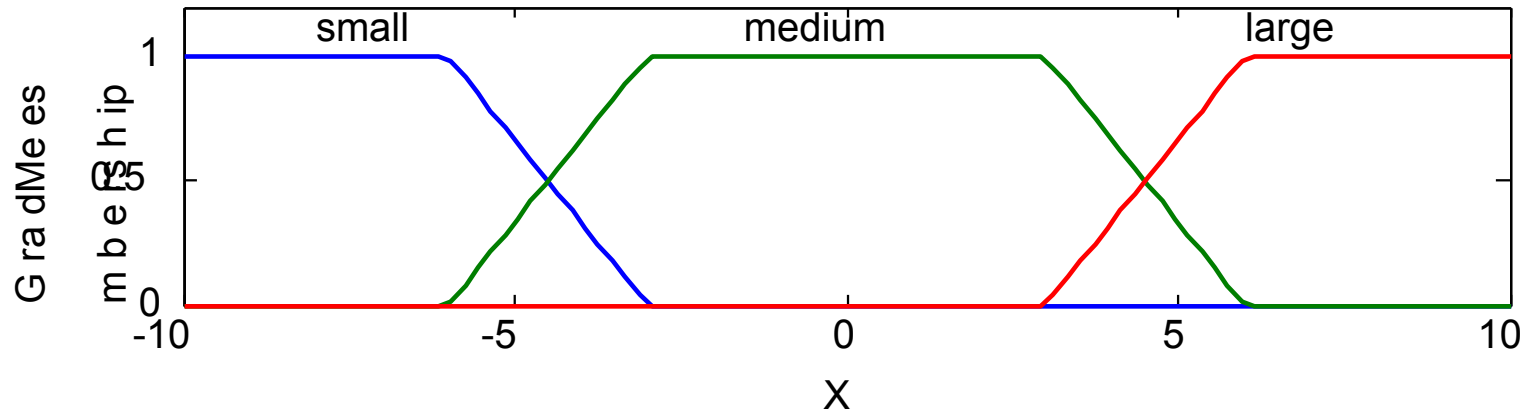
Mamdani composition of two-rule fuzzy system
max-min inferencing

<http://aragorn.pb.bialystok.pl/~radev/logic/logrozm.htm>

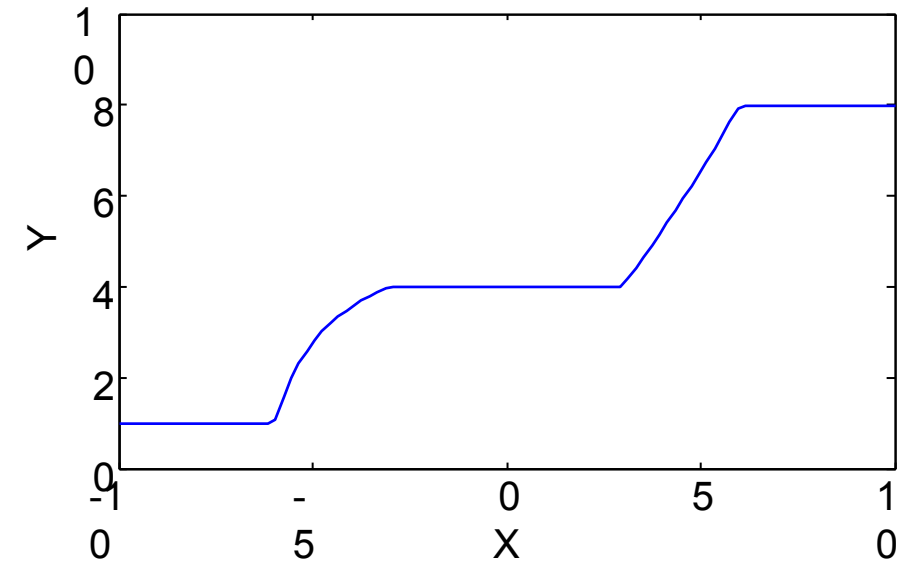
Defuzzification



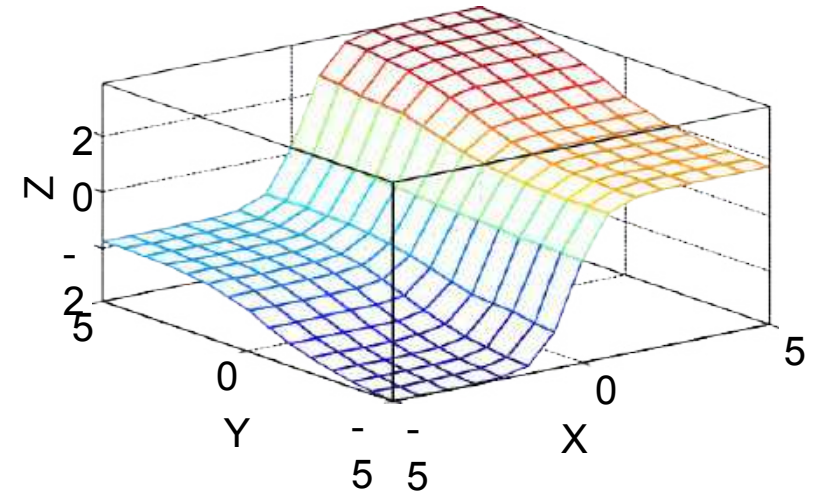
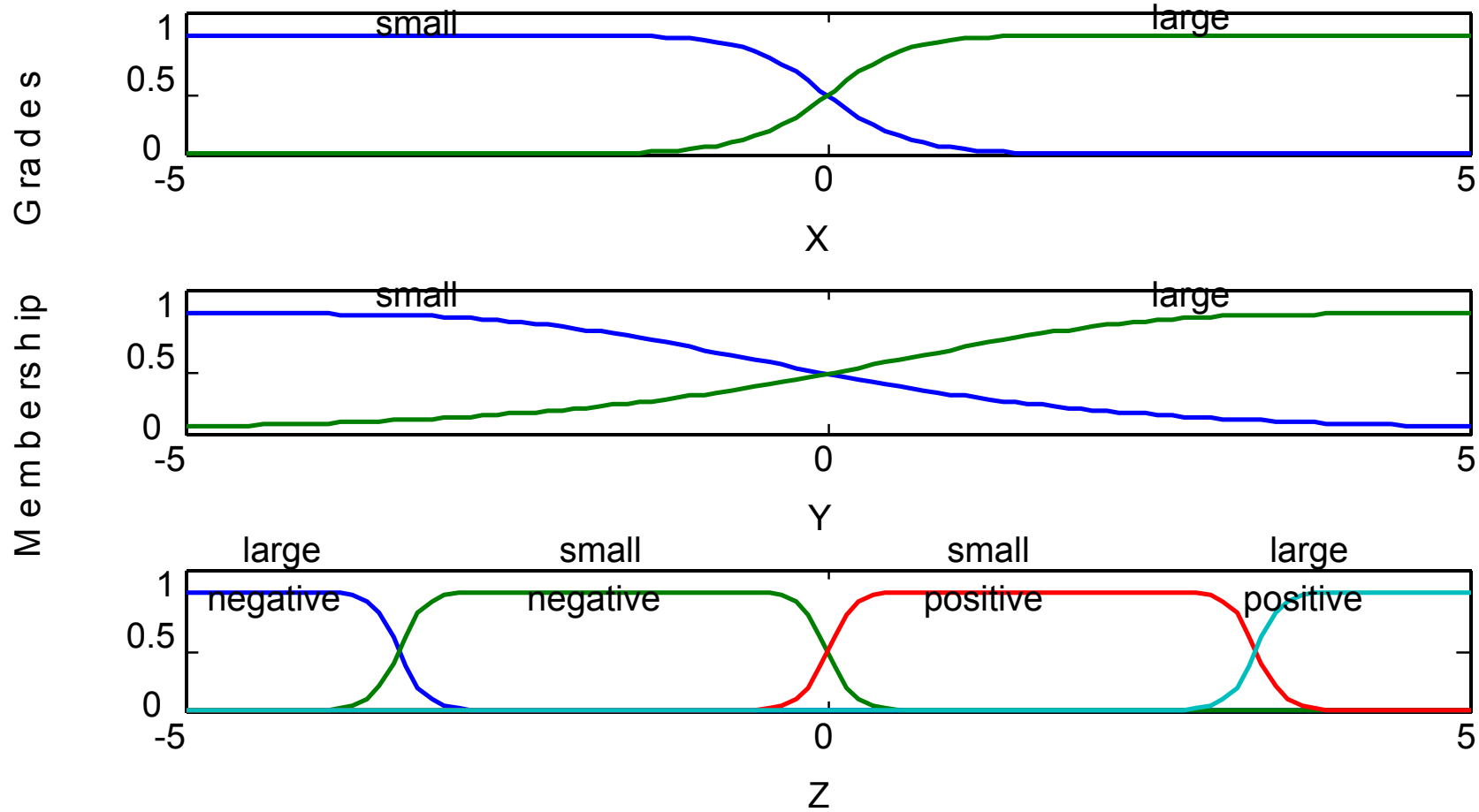
Jang, page 77. See pages 75–76 for formulas.



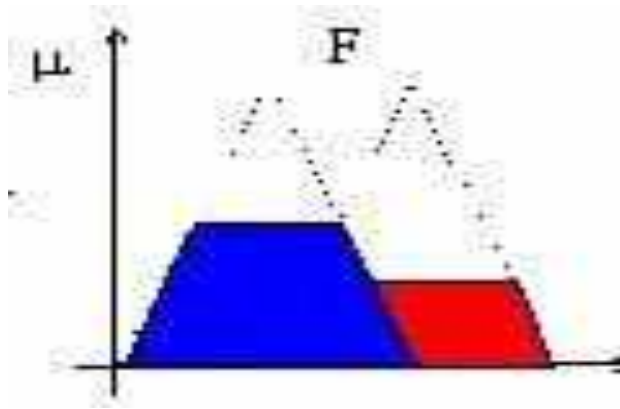
- if x is small then y is small
- if x is medium then y is medium
- if x is large then y is large



Results depend on T- norm, S-norm, and defuzzification method



- if x is small & y is small then z is large negative
- if x is small & y is large then z is small negative
- if x is large & y is small then z is small positive
- if x is large & y is large then z is large positive



Defuzzification can be computationally expensive. What is the center of area of this fuzzy set?

Theorem 4.1: If we use the product T-norm, and the summation S-norm (which is not really an S-norm), and centroid defuzzification, then the crisp output z is:

$$z = \frac{w_i a_i z_i}{w_i a_i}$$

w_i = firing strength (input MF value)

a_i = consequent MF area

z_i = consequent MF centroid

a_i and z_i can be calculated ahead of time!

Sugeno Fuzzy Model

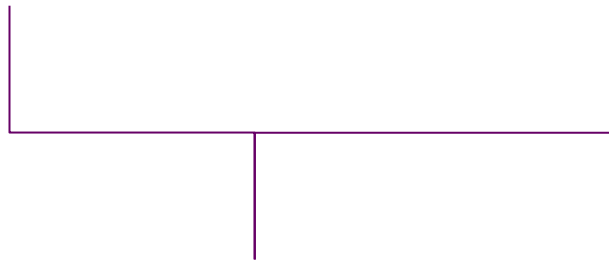
- The Sugeno FIS is quite similar to the Mamdani FIS. The primary difference is that the output consequence is not computed by clipping an output membership function at the rule strength.
- In fact, in the Sugeno FIS there is no output membership function at all.
- Instead, the output is a crisp number computed by multiplying each input by a constant and then adding up the results.

Zero-order TSK Fuzzy Model

- When f is constant, we have a zero-order TSK fuzzy model (a special case of the Mamdani fuzzy inference system which each rule's consequent is specified by a fuzzy singleton or a pre defuzzified consequent)
- Minimum computation time

Fuzzy Rules of TSK Model

If x is A and y is B then $z = f(x, y)$



Fuzzy Sets



Crisp Function

$f(x, y)$ is very often a polynomial
function

Examples

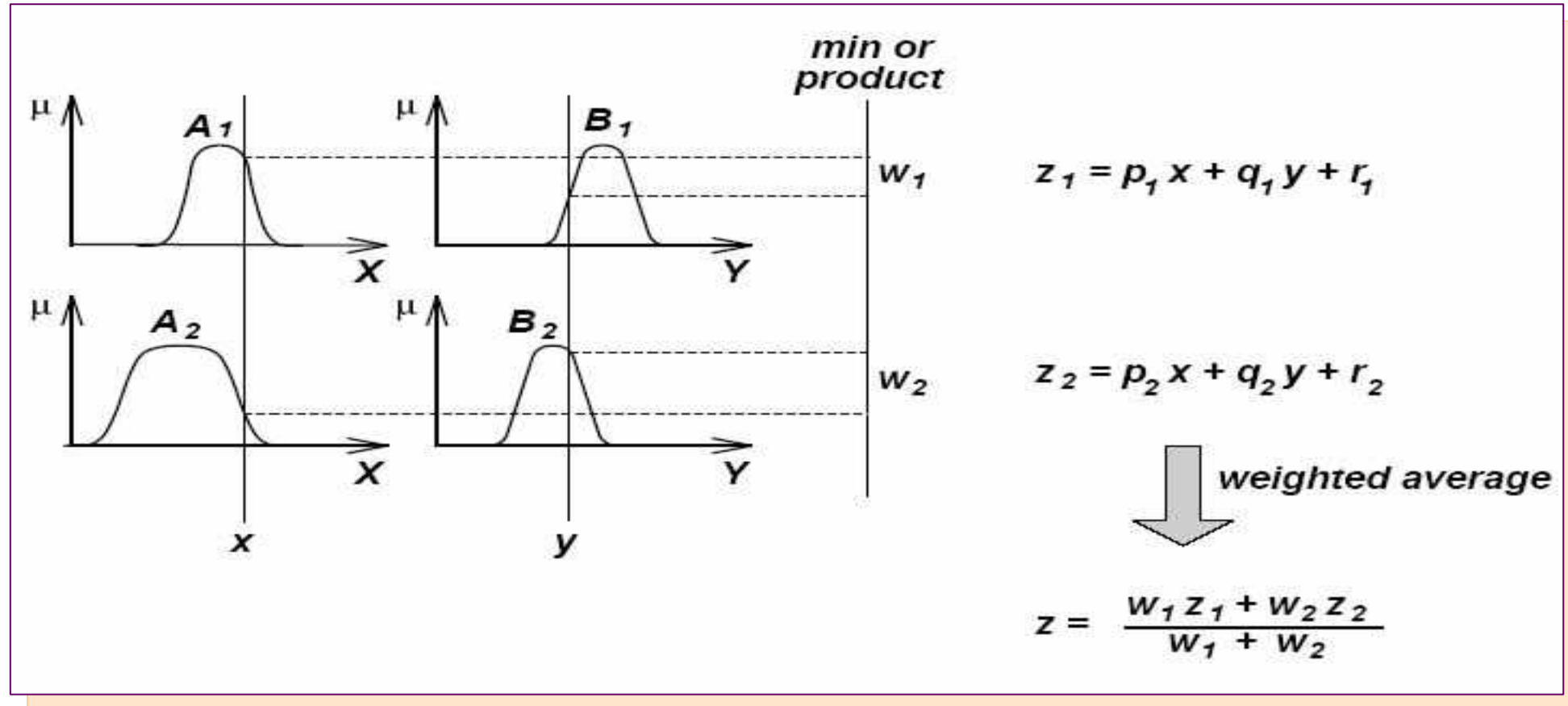
R1: if X is small and Y is small then $z = -x + y + 1$

R2: if X is small and Y is large then $z = -y + 3$

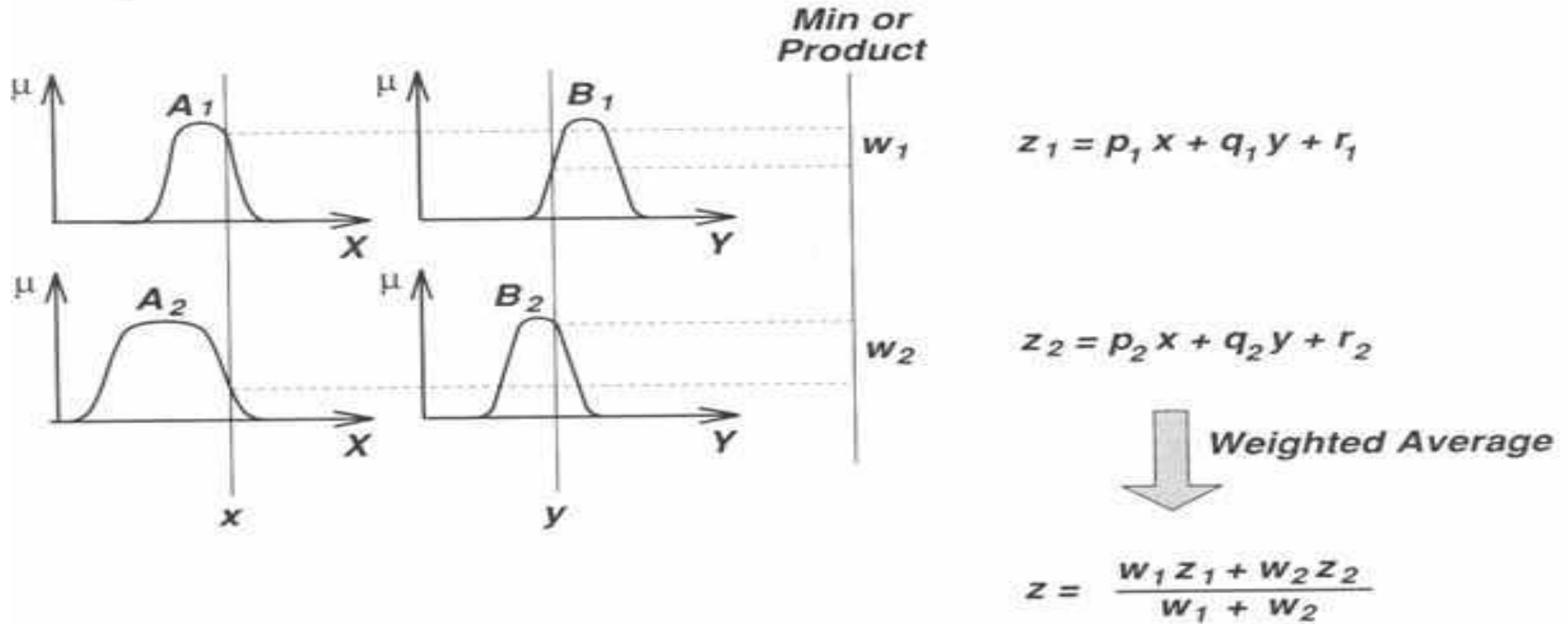
R3: if X is large and Y is small then $z = -x + 3$

R4: if X is large and Y is large then $z = x + y + 2$

The Reasoning Scheme



First-Order: TSK Fuzzy Model



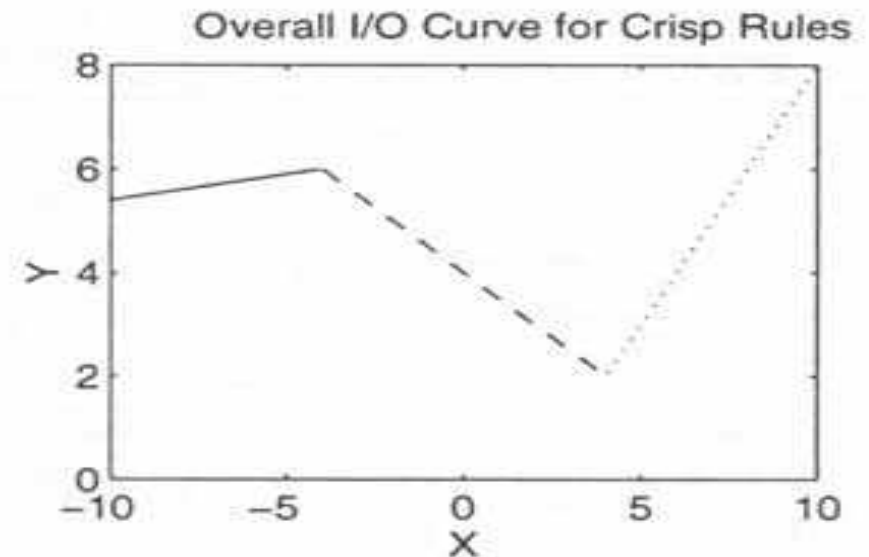
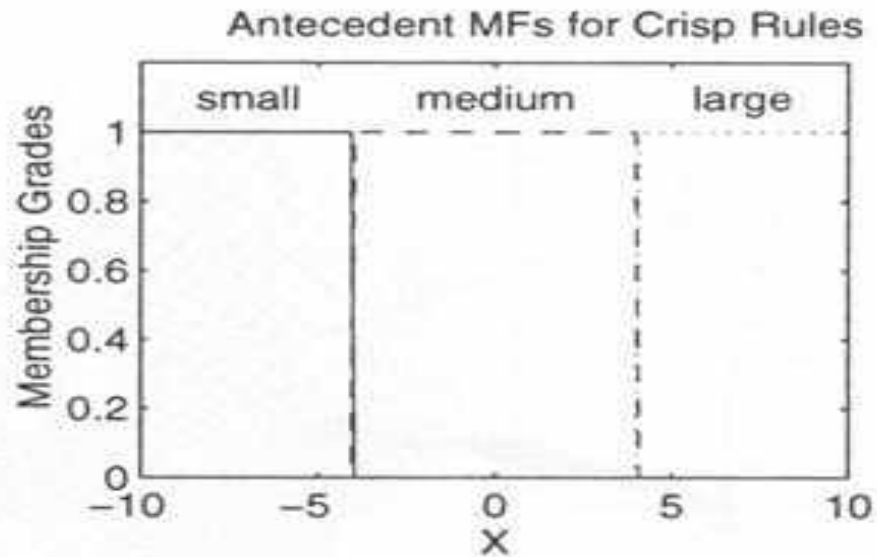
Example #1: Single-input

- **A single-input TSK fuzzy model can be expresses as**

- **If X is small then $Y = 0.1X + 6.4$.**
- **If X is medium then $Y = -0.5X + 4$.**
- **If X is large then $Y = X - 2$.**

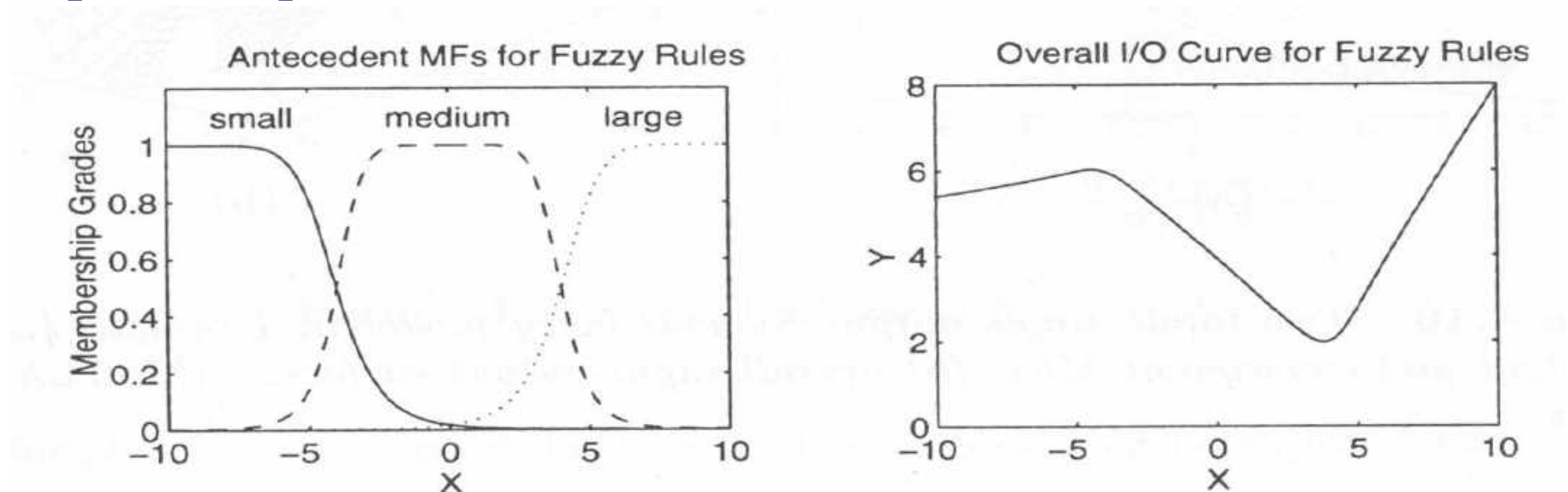
Example #1: Non fuzzy rule set

- If “small”, “medium.” and “large” are non fuzzy sets , then the overall input-output curve is piecewise



Example #1: Fuzzy rule set

- If “small”, “medium,” and “large” are fuzzy sets (smooth membership functions), then the overall input-output curve is a smooth one.

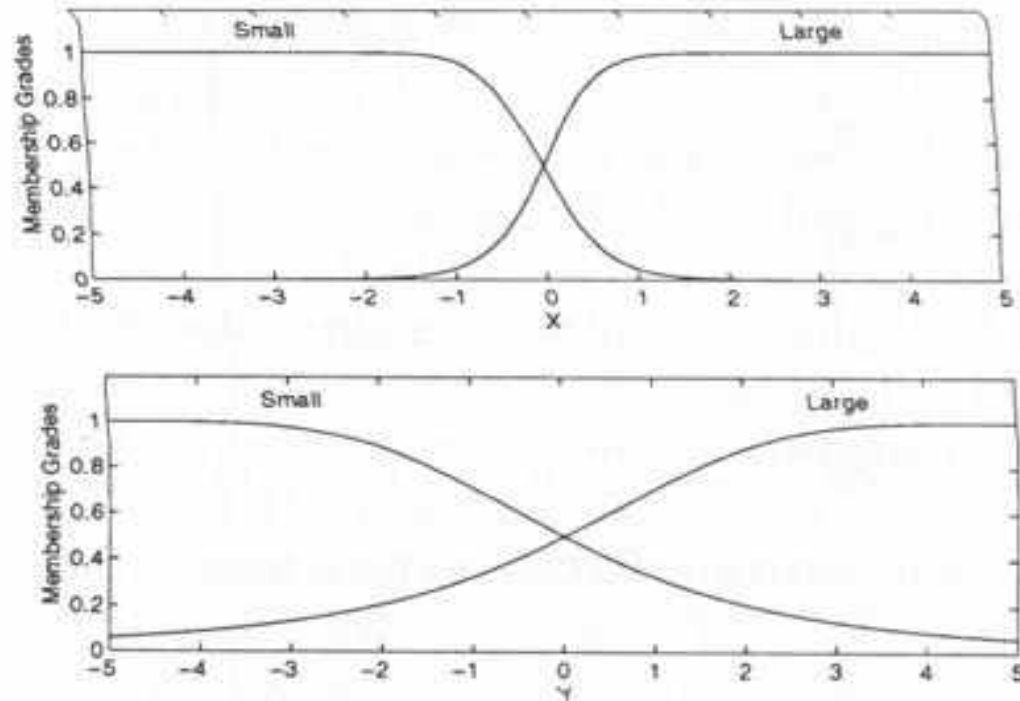


Example #2 : Two-input

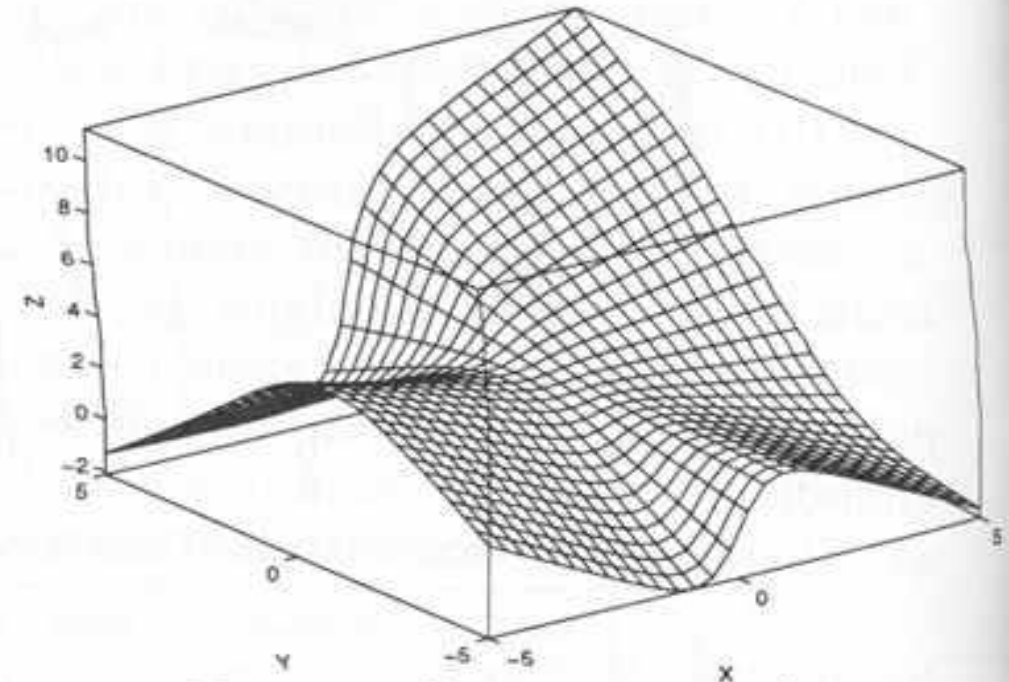
A two-input TSK fuzzy model with 4 rules can be expressed as

- If **X is small** and **Y is small** then **$Z = -X + Y + 1$.**
- If **X is small** and **Y is large** then **$Z = -Y + 3$.**
- If **X is large** and **Y is large** then **$Z = X - Y + 2$.**
- If **X is large** and **Y is small** then **$Z = X + Y + 3$.**

Example #2 : Two-input



Input X and Y MFs



Input-Output surface

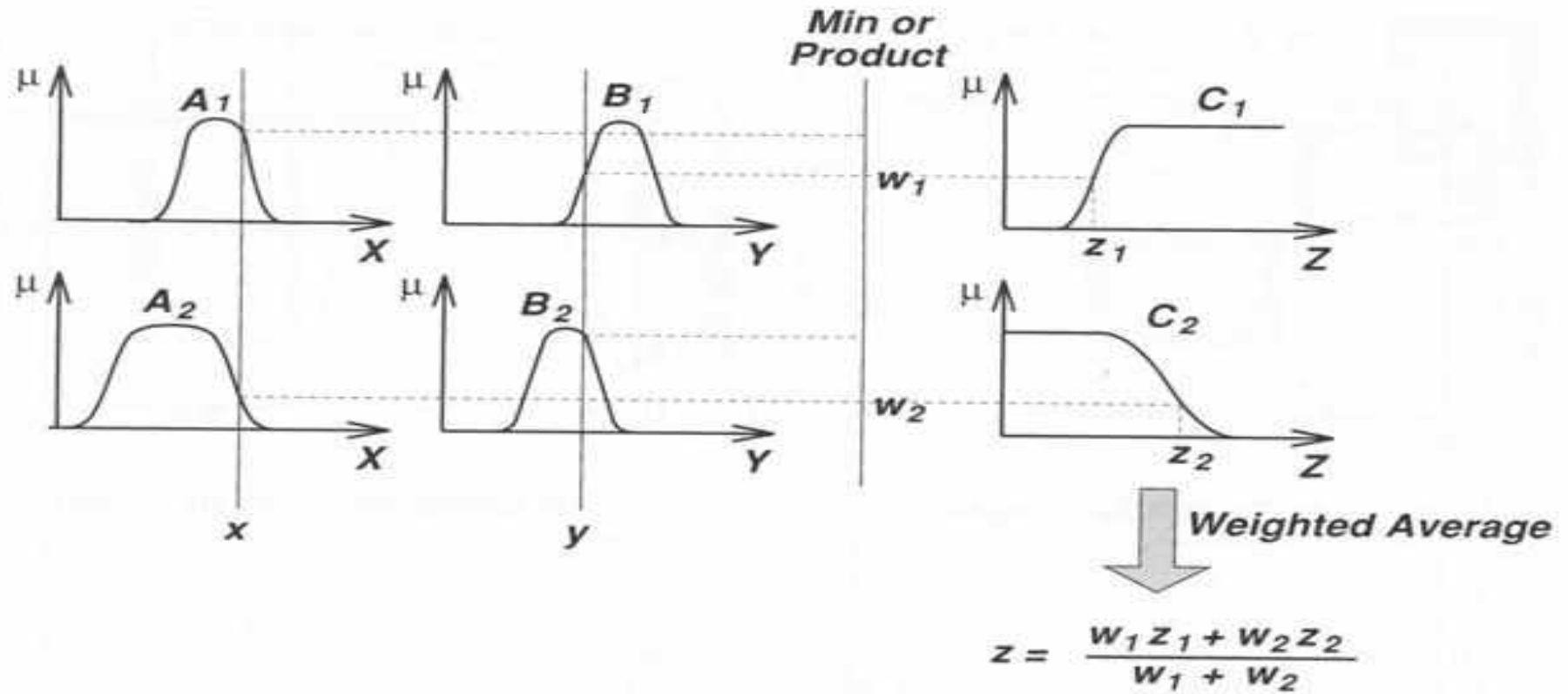
Summary: TSK Fuzzy Model

- Overall output via either weighted average or weighted sum is always crisp
- Without the time-consuming defuzzification operation, the TSK (Sugeno) fuzzy model is by far the most popular candidate for sample-data-based fuzzy modeling.
- Can describe a highly non-linear system using a small number of rules
- Very well suited for adaptive learning.

Tsukamoto Fuzzy Models

- The consequent of each fuzzy if-then rule is represented by a fuzzy set with **monotonical MF**
 - As a result, the inferred output of each rule is defined as a crisp value induced by the rules' firing strength.
- The overall output is taken as the weighted average of each rule's output.

Tsukamoto Fuzzy Models



Example: Single-input Tsukamoto fuzzy model

- **A single-input Tsukamoto fuzzy model can be expressed as**

- **If X is small then Y is C1**
- **If X is medium then Y is C2**
- **If X is large then Y is C3**

Tsukamoto Fuzzy Models

The rule base has the form as:

R_i : if u is A_i and v is B_i , then w is C_i , $i = 1, 2, \dots, n$

where $\mu_{C_i}(w)$ is a monotonic function.

As a result, the inferred output of each rule is defined as a crisp value induced by the rule's matching degree (firing strength). The overall output is taken as the weighted average of each rule's output.

We suppose that the set C_i has a monotonic membership function $\mu_{C_i}(w)$ and that α_i is the matching degree of i th rule.

(1) For the singleton input (u_0, v_0)

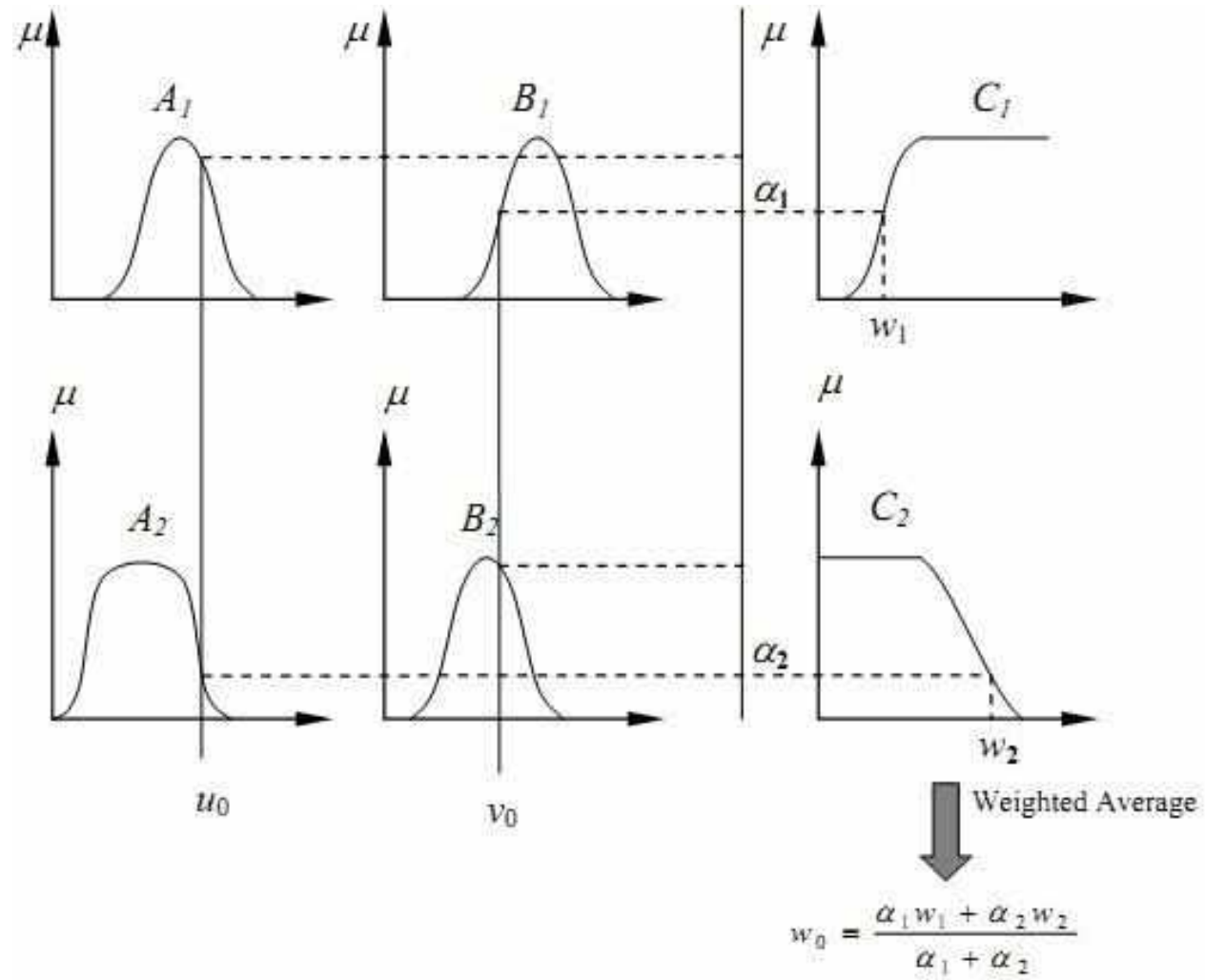
$$\alpha_i = \mu_{A_i}(u_0) \wedge \mu_{B_i}(v_0)$$

(2) For the fuzzy set input (A', B')

$$\alpha_i = \min\left[\max_u(\mu_{A'}(u) \wedge \mu_{A_i}(u)), \max_v(\mu_{B'}(v) \wedge \mu_{B_i}(v))\right]$$

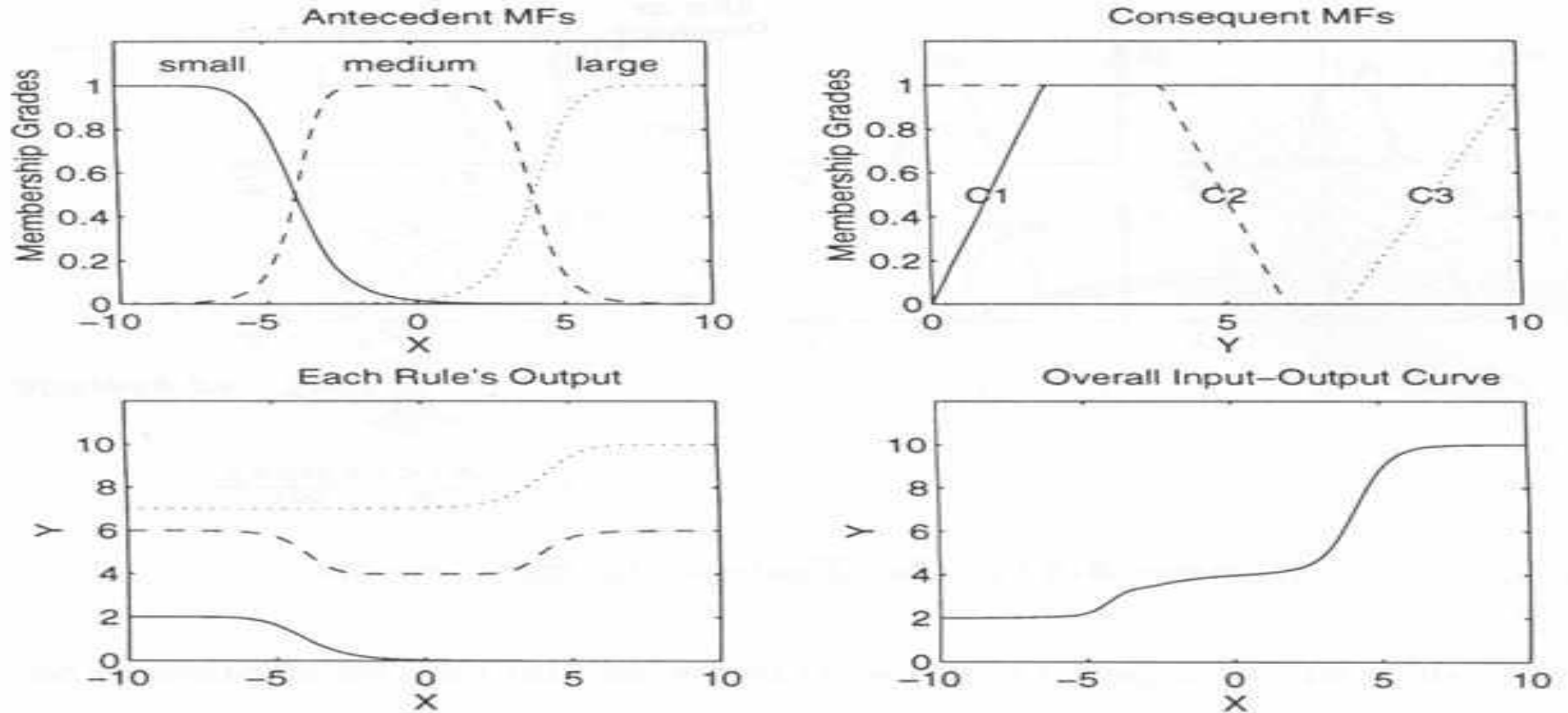
Then the result of i th rule is obtained by

Tsukamoto Fuzzy Models



Graphical representation of Tsukamoto method

Example: Single-input Tsukamoto fuzzy model



Applications of Fuzzy Logics

- Consumer Electronics
- Automotive Systems
- Industrial Automation and Control
- Decision-Making and Expert Systems
- Image Processing and Computer Vision
- Natural Language Processing
- Home Automation & Smart Devices
- Agriculture and Environment
- Transportation Systems
- Financial and Economic Systems